DD-G-1403C
September 13, 1983
SUPERSEDING
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August 15, 1972

FEDERAL SPECIFICATION

GLASS, FLOAT, SHEET, FIGURED, COATED (HEAT-STRENGTHENED AND TEMPERED)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heat strengthened and tempered flat glass used in general building construction.

1.2 Classification.

1.2.1 Grades. Glass furnished under this specification shall be of the following grades, as specified (see 6.2):

Grade A - Heat-strengthened glass.
Grade B - Tempered glass.

1.2.2 Styles. Glass shall be of the following styles, as specified (see 6.2):

Style I - Uncoated surfaces.
Style II - One surface ceramic coated.
Style III - Coated surface, transparent coating.

1.2.3 Styles, types, classes, kinds, forms, qualities, finishes, and patterns. Glass shall be of the following styles, types, classes, kinds, forms, qualities, finishes, and patterns, as specified (see 6.2):

Style I, uncoated surfaces:

Type I - Primary glass (float or sheet).
Quality q3 - Glazing select.
Quality q5 - Glazing B.
Class 1 - Transparent.
Class 2 - Heat absorbing and light reducing.

Kind A - Higher light transmittance.
Kind B - Lower light transmittance.

Class 3 - Light reducing, tinted.
Class 4 - Translucent (see 3.1.6).

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Fine preparation for delivery requirements

Type II - Rolled flat.
Quality q7 - Decorative.
Quality q8 - Glazing.

Class 1 - Translucent.
Class 2 - Heat absorbing.

Kind A - Higher light transmittance.
Kind B - Lower light transmittance.

Class 3 - Light reducing, tinted.
Form 3 - Figured.

Finish 1 - Figured one side.
Finish 2 - Figured both sides.

Pattern p1 - Linear.
Pattern p2 - Geometric.
Pattern p3 - Random.
Pattern p4 - Special.
Pattern p12 - Custom designs or textures (see 3.12.4).

Style II, one surface ceramic coated.

Type I - Primary glass (float or sheet).

Quality q3 - Glazing select.
Quality q5 - Glazing B.

Type II - Rolled flat.

Form 3 - Figured.

Quality q7 - Decorative.
Quality q8 - Glazing.
Finish f6 - Finish f1 with either side ceramic coated.

Type III - Style III glass.

Style III, coated surface, transparent coating.

Type I - Primary glass (float or sheet).

Quality q3 - Glazing select.

Class 1 - Transparent.
Class 2 - Heat absorbing and light reducing.

Kind A - Higher light transmittance.
Kind B - Lower light trasmittance.

Class 3 - Light reducing, tinted.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

DD-G-451 - Glass, Float or Plate, Sheet, Figured (Flat, for Glazing, Mirrors and Other Uses).
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).

Federal Regulation:

Consumer Product Safety Commission (CPSE)

16 CFR Part 1201 - Safety Standard for Architectural Glazing Materials.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

BUILDING OFFICIALS & CODE ADMINISTRATORS INTERNATIONAL, INC. (BOCA):

The BOCA Basic Building Code

(Application for copies should be addressed to the Building Officials & Code Administrators International, Inc., 1313 East 60th Street, Chicago, Illinois 60637).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description.

3.1.1 Grade A. Grade A glass shall be plate, float, sheet, or rolled glass, hereinafter specified as heat strengthened glass.

3.1.2 Grade B. Grade B glass shall be plate, float, sheet, or rolled glass, hereinafter specified as tempered glass.

3.1.3 Style I, uncoated glass. Style I glass shall be heat strengthened or tempered with no coatings or special surface treatment. Surface designs, finishes, patterns, and textures provided by sandblasting, etching, or rolling to meet fabrication requirements in 3.12.1 shall not be considered to be special surface treatments.

3.1.4 Style II, one surface ceramic coated glass. Style II glass shall be heat strengthened or tempered that has been coated on one side with ceramic coating which has been fused onto and made an integral part of the surface of the glass.

3.1.5 Style III, coated surface, transparent coating. Style III glass shall be heat-strengthened or tempered that has been coated on one or both sides with a transparent coating before, during or after the heat strengthening or fully tempering operation. Type I transparent coating produced in a range of colors or tints, transmittance reflectance, and qualities shall be the manufacturer's standard commercial product.

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3.1.6 Type I, class 4. Float or sheet glass made translucent overall by sandblasting, etching or similar methods (see 1.2.3).

3.2 Standard commercial product. The glass shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the glass being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Flatness.

3.4.1 Localized warp. Localized warp for rectangular glass shall not exceed 1/16 inch over any 12-inch span. Localized warp for strips shall not exceed 3/32 inch over any 12-inch span (see 3.5).

3.4.2 Localized kink. Any localized kink centered at any tong location shall not exceed 1/16 inch in a 2-inch span.

3.4.3 Overall bow and warpage tolerances. Overall bow and warpage tolerances shall not exceed the deviations from a flat surface shown in table I.

TABLE I. Overall bow and warpage tolerances, maximum.

	Glass thickness inches				
Length of span inches	1/8*, 5/32 3/16, .210	7/32, 1/4	5/16	3/8	1/2 and Over
up to 36	0.188(.094)	0.235	0.109	0.093	0.063
36 to 48	0.282(.125)	0.188	0.171	0.141	0.093
48 to 60	0.374(.156)	0.250	0.219	0.187	0.141
60 to 72	0.470(.188)	0.375	0.281	0.235	0.187
72 to 84	0.561(.219)	0.500	0.328	0.282	0.235
84 to 96	0.940(.250)	0.625	0.437	0.375	0.282
96 to 108	-----	0.750	0.562	0.500	0.375
108 to 120	-----	0.875	0.687	0.625	0.500
120 to 132	-----	1.000	0.812	0.750	0.625

* Tolerances shown in parentheses apply to 1/8-inch thickness only when the alternate checking procedures in 4.3.4 are used.

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3.5 Strips, style I, II, III glass. Strips shall be determined as follows:

Minimum width of a strip to comply with strip tolerances shall be equal to or greater than the length divided by 10.

Up to 72 inches long - When the width is equal to or less than the length divided by 8.

Over 72 inches to 96 inches long - When the width is equal to or less than the length divided by 7.

Over 96 inches to 132 inches long - When the width is equal to or less than the length divided by 5.

3.6 Strength. The strength of style I, II, and III glass shall be tested in accordance with 4.4.2. The working strength of glass for any location in which it is required to withstand wind or impact loads (see 3.6.4 and 4.4.3.6) shall be determined according to the design procedure and criteria of BOCA Basic Building Code.

3.6.1 Grade A, heat-strengthened glass. When tested in accordance with 4.4.2.1, grade A glass shall satisfy either a. or b:

a. A surface compression of not less than 3,500 pounds per square inch (lb/in2) and not more than 10,000 lb/in^2.

b. An average edge compression of not less than 5,500 lb/in^2.

3.6.2 Grade B, tempered glass. When tested in accordance with 4.4.2.1, grade B glass shall satisfy either a. or b:

a. A surface compression of not less than 10,000 lb/in2.
b. An average edge compression of not less than 9,700 lb/in^2.

3.6.3 Uniform load strength, grade A glass. When tested in accordance with 4.4.2.2, grade A glass shall meet the requirements of table II.

TABLE II. Uniform load strength, grade A, heat-strengthened glass.

Glass thickness (inch)	Average breaking pressure X glass area. (minimum value) *
1/8	1,325
5/32	2,000
3/16	3,325
.210	3,325
7/32	3,325
1/4	5,400
5/16	7,350
3/8	12,000
1/2	16,000

* Average breaking pressure in pounds per square foot and area in square foot.

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3.6.4 Impact strength, grade B glass. When specified (see 6.2), the impact strength of grade B glass shall be tested and interpreted in accordance with 4.4.2.3.

3.7 Fallout resistance. When specified (see 6.2), style II glass for installation above areas used by the public, such as sidewalks, pedestrian or vehicular ramps, paved plazas and entrances shall meet the fallout resistance when tested in accordance with 4.4.3.

3.8 Color or tint, style I, II, and III glass. The color or tint for each grade, style, type, class, kind, finish, or pattern shall be as specified (see 6.2).

3.9 Transmittance, style I, class 2, kind A and B. In addition to the requirements of this specification, style I, class 2, kind A and B heat absorbing glass shall be in accordance with the requirements for illuminant C transmission and solar transmittance in DD-G-451 (see 4.4.4).

3.10 Luminous transmittance, style I, class 3. Light reducing glass is produced in a wide range of luminous transmittance values. The luminous transmittance shall be as specified (see 6.2).

3.11 Identification marking. The identification marking shall include the manufacturer's name or trademark, the glass thickness, and the designation of heat strengthened or fully tempered, as applicable. The identification marking shall be either permanent or temporary, as required. Glass intended for use in vision areas (such as doors and windows) shall be permanently marked. Glass intended for use in nonvision areas (such as spandrels) shall have either permanent or temporary identification marking at the manufacturer's option, unless otherwise specified (see 6.2).

3.11.1 Permanent identification marking. Permanent identification marking shall be accomplished by sandblasting, etching, or fired-on ceramic decal or silk screened imprint that do not introduce stresses. The marking shall be located at a corner of the glass where it shall be fully visible and legible when the glass is installed.

3.11.2 Temporary identification marking. Temporary identification marking shall consist of a temporary label attached to the glass which can be removed after installation.

3.12 Workmanship.

3.12.1 Fabrication. All fabrication, such as cutting to overall dimensions, edgework, drilled holes, notching, grinding, sandblasting, and etching shall be performed prior to strengthening or tempering and shall not be modified except as recommended by the fabricator. No modification shall be made which will affect its performance characteristics as herein specified.

3.12.2 Fittings and hardware. Requirements for fittings and hardware shall be as specified (see 6.2), or as shown on plans or drawings. Fittings and hardware specified shall be compatible with glass fabrication limitations.

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3.12.3 Tong marks. The center of tong marks, when present, shall be located a maximum of 1/2 inch from one edge of the glass. Tong marks shall be located on a specific edge when specified (see 6.2). For location of tong marks on glass with special fabrication or irregular patterns, consult fabricators.

3.12.4 Pattern p12, type II. Custom designs or textures shall be as specified (see 6.2), or as shown on plans or drawings.

3.12.5 Glass quality and finish. Glass qualities q3, q5, q7, and q8 in this specification shall be in accordance with the applicable requirements for qualities q3, q5, q7, and q8 in DD-G-451.

3.13 Thickness. Heat-strengthened and tempered glass shall be in accordance with the thickness requirements of Table III and IV. Sizes and thicknesses of grade A and B glass are subject to the processing limitations of each furnace.

Table III: Thickness and Thickness Tolerance for Float or Sheet Glass, Style I, II, and III, Type I

Thickness designation		Thickness tolerance (inches)			
		Style I and III		Style II	
inches	mm	Min.	Max.	Min.	Max.
1/8	3.0	0.115	0.134	0.115	0.137
5/32	4.0	0.149	0.165	0.149	0.168
3/16	5.0	0.180	0.199	0.180	0.202
7/32	5.5	0.200	0.218	0.200	0.221
1/4	6.0	0.219	0.244	0.219	0.247
5/16	8.0	0.292	0.332	0.292	0.335
3/8	10.0	0.355	0.406	0.355	0.409
1/2	12.0	0.469	0.531	0.469	0.534
5/8	16.0	0.594	0.656	0.594	0.659
3/4	19.0	0.719	0.781	0.719	0.784
7/8	22.0	0.844	0.906	0.844	0.909
1	25.0	0.969	1.031	0.969	1.034
1-1/4	32.0	1.125	1.375	1.125	1.378

Table IV: Thickness and Thickness Tolerance for Rolled Glass, Type II, Form 3

Style	Finish	Thickness Designation		Tolerance (Inches)	
		Inches	mm	Min	Max.
I	f1 & f2	1/8	3.2	0.109	0.172
		5/32	4.0	0.140	0.203
		3/16	4.8	0.171	0.319
		0.210	5.3	0.194	0.241
		7/32	5.6	0.203	0.266
		1/4	6.4	0.234	0.297
		5/16	7.9	0.281	0.359
		3/8	9.5	0.343	0.422
II	f6	7/32	5.6	0.203	0.269
		5/16	7.9	0.281	0.362

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All glass offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.2.1 Sampling for examination. Examination shall be based on inspection level S-4 and an Acceptable Quality Level (AQL) of 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.2.2 Sampling for tests. Tests shall be based on inspection level S-1 and an AQL of 4.0 percent defective.

4.3 Examination. Each sample selected shall be examined for defects listed in table V and for the requirements of 4.3.1 through 4.3.4.

TABLE V. Classification of defects.

Classification	Defects	Requirement paragraph
Major:		
101	Classification not as specified.	1.2 and 3.1
102	Localized warp exceeds requirements.	3.4.1
103	Localized kink exceeds requirements.	3.4.2
104	Overall bow and warpage exceed tolerances.	3.4.3
105	Color or tint not as specified.	3.8
106	Fabrication not as specified.	3.12.1
107	Fittings and hardware not as specifed.	3.12.2
108	Glass quality and finish not as specified.	3.12.5
Minor:		
201	Identification marking not as specified.	3.11
202	Tong marking location not as specified.	3.12.3

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4.3.1 Wave, ream, strings, and lines, style I and III, type 1. Place sample in a vertical position at a distance of approximately 36 inches from a brick wall or similar background showing essentially straight lines. The viewer shall look through direct sunlight or with background light suitable for observing each type of defect. View the sample at an angle to the surface of not less than the vision interference only in the table for the applicable glass in DD-G-451. Line of vision shall be perpendicular to the wall. Acceptability shall be determined in accordance with the applicable table of defects in DD-G-451.

4.3.2 Scratches, rubs, stones, and gaseous inclusions, style 1, types 1, and 11, and style III, type 1. Place sample in a vertical position approximately 36 inches from the viewer's position. The viewer shall look through the sample using daylight without direct sunlight or with background light suitable for observing each type of defect. Acceptability shall be determined in accordance with the applicable defects in DD-G-451 and 3.12.5.

4.3.3 Defects and appearance, style 11. One surface ceramic coated glass shall be viewed by light reflected from the viewed surface. Place the glass sample against opaque backup material and in a vertical position. View the sample from a distance of approximately 36 inches for glass defects and determine acceptability in accordance with the applicable defects in DD-G-451 and 3.12.5.

4.3.4 Localized warp and overall bow and warpage. Place sample glass in a free standing vertical position, resting on blocks at the quarter points. With the glass in this position, place a straight edge across the concave surface parallel to and within 1/2 inch of the edge and measure the maximum deviation with a taper or feeler gage, or a dial indicator may be used. When the above procedure is impractical for larger sizes of 1/8-inch thickness, the glass may be placed on a flat surface concave side down, and a dial indicator, reading in 0.001-inch increments, shall be used to determine overall bow and warpage. Overall bow and warpage tolerances shown in parentheses in table 1 shall apply when the alternate procedure is used (see 3.4 and table 1).

4.4 Tests.

4.4.1 Specimens. The number of specimens required shall be selected from each sample selected. Two specimens shall be required for testing except as shown in a, b, or c.

a. Uniform load strength test requires eight specimens (see 4.4.2.2).
b. Fallout resistance test of style 11 glass required three specimens (see 4.4.3).
c. Impact test of grade B glass requires four specimens (see 4.4.2.3). Due to the limitations of sizes which fit in the impact frame, specimens may be either production glass or special specimens, within the requirements of CPSC 16CFR1201, prepared and processed with the production glass.

4.4.2 Strength requirements, grade A and B. Determine strength requirements of grade A glass in accordance with either 4.4.2.1 or 4.4.2.2 at the manufacturer's option.

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4.4.2.1 Surface and edge compression, grade A and B. The strain pattern in which edge and surface compression can be determined shall be measured by the polariscope method. When the range of the apparatus permits examination for edge compression only, the average value for the midpoints of each edge shall be determined. Style II glass shall have the ceramic coating removed to accomplish this examination. The ceramic coating may be removed with hydrofluoric acid or abrasive cloth. Index oil and a glass slide cover plate may be necessary to eliminate the abraded surface to see compression color bands (see 3.6.1 and 3.6.2).

4.4.2.2 Uniform load strength, grade A glass. Subject eight samples of glass to uniform load test as follows. All samples shall be the same size and thickness. The size of the test samples shall be 36 by 72 inches or at least 75 percent of the actual size in each dimension, if actual size is smaller than 36 by 72 inches (see 3.6.3).

a. Mount glass in opening in vacuum chamber so glass is supported on all four edges and is prevented from touching any metal or other hard surfaces by a minimum of 1/8 inch soft material, such as neoprene rubber.
b. Reduce the pressure in the chamber at a rate sufficient to produce 0.1 inch deflection each minute. These deflections may be made uniformly or in increments of 0.1 inch each minute.
c. Record the breaking pressures and average for the eight samples.

4.4.2.3 Impact strength, grade B, tempered glass. Test and interpret in accordance with CPSC 16CFR1201 (see 3.6.4).

4.4.3 Fallout resistance test, style II glass.

4.4.3.1 Frequency of tests. Unless otherwise specified (see 6.2), the test for fallout resistance shall be performed on the initial inspection, and thereafter only when changes are made in the material.

4.4.3.2 Specimen size. Size of specimens shall be 34 by 76 inches with a tolerance of +1/16 inch - 3/16 inch on each dimensions.

4.4.3.3 Mounting. Mount glass in a frame in accordance with CPSC 16CFR1201.

4.4.3.4 Test procedure. Test for 100 cycles, each cycle to proceed in sequence a, b, c, and d as follows and repeat with no time delay between cycles:

a. One hour at 0° Fahrenheit (F).
b. During the next 3 hours increase temperature from 0° to 140°F, with relative humidity at 95 to 100 percent above 40°F.
c. Then hold for 1 hour at 140°F, 95-100 percent relative humidity; and
d. During the next 3 hours, decrease temperature from 140°F, ambient humidity.

4.4.3.5 Fracturing glass. Break glass using a spring-loaded prick punch at the midpoint of either vertical edge. Cracks and fissures which may develop are permissible (see 3.6).

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4.4.3.6 Wind load. After glass has been broken, subject it to a pressure of 4 pounds per square foot for 5 minutes to simulate action of wind load against a building (see 3.6).

4.4.3.7 Interpretation of tests. Although cracks and fissures which may be developed are permissible, no opening shall occur through which a 3-inch sphere may be freely passed.

4.4.4 Luminous and solar transmittance. Test for luminous and solar transmittance in accordance with DD-G-451 (see 3.9 and 3.10). Changes in transmittance from the stated value are acceptable if the percent transmission is within the stated value ±10.

4.5 Preparation for delivery inspection. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one unit prepared for shipment. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.1.1 Level A. Glass of like description in quantities as specified (see 6.2), shall be packed in a close-fitting box conforming to PPP-B-621, class 2. Each pane of glass shall be separated with a sheet of neutral or slightly acid newsprint or kraft paper of equal dimensions to the glass or powder separator, -1/4 inch, or of greater dimensions than the glass. The boxes shall be lined on all interior surfaces with not less than three layers of fiberboard conforming to PPP-F-320, type CF, class domestic, variety SW.

5.1.2 Level B. The glass shall be packed as specified in 5.1.1 except that containers shall be of a domestic type. Fiberboard containers conforming to PPP-B-636, class weather resistant, and PPP-B-640, class 2, may also be utilized.

5.1.3 Commercial. Glass shall be packed to insure carrier acceptance and safe delivery to destination in compliance with carrier rules and regulations.

5.2 Marking.

5.2.1 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

5.2.2 Military activities. Shipments to military activities shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Grade A, style 1, type I and II, and style III, type I. Intended for glazing where additional strength and better resistance to thermal breakage is desired, but not requiring the safety break properties or strength of fully tempered glass.

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6.1.2 Grade B, style I, types I, II, and style III, type I. Intended for general glazing, sliding doors, storm doors, building entrances, bath and shower enclosures, counter tops, showcases, interior partitions, and other uses where the superior strength or safety properties of fully tempered glass are required.

6.1.3 Style I, class 2. Intended for use where reduction of solar heat is desired in addition to fully tempered or heat strengthened properties.

6.1.4 Style I, class 3. Intended for use where reducing visible light transmission and glare is desired, in addition to fully tempered or heat strengthened properties.

6.1.5 Grade A or B, style II, types I and II. Intended for use in nonvision glass areas with opaque backup construction, and where no light source originates behind the ceramic coated glass surfaces.

6.1.6 Style III. Intended for use where reduction of solar heat, loss of internal heat and/or light transmission is desired in addition to fully tempered or heat strengthened properties.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Grade, style, type, class, kind, form, quality, finish, and pattern of glass (see 1.2).
c. When impact strength test is required for grade B glass (see 3.6.4).
d. Style II glass, installation (see 3.7).
e. Color or tint of glass (see 3.8).
f. Luminous transmittance of style I, class 3 glass (see 3.10).
g. When either permanent or temporary identification marking is required (see 3.11).
h. Requirements for fittings and hardware (see 3.12.2).
i. Specific location of tong marks, when required (see 3.12.3).
j. Custom design or texture required (see 3.12.4).
k. Style II glass, frequency of test for fallout resistance (see 4.4.3.1).
l. Level of packing required (see 5.1).
m. Quantities of glass required in shipping container (see 5.1.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.4 Cross reference data. Cross reference of classification changes between this specification and the preceding issue DD-G-1403B are as follows:

DD-G-1403B	DD-G-1403C
Condition A, uncoated surfaces:	Style 1, uncoated surfaces:
Kind HS - Heat strengthened glass	Grade A - Heat strengthened glass
Kind FT - Fully tempered glass	Grade B - Tempered glass

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Condition A - Uncoated surfaces
Condition B - Spandrel glass, one surface ceramic coated

Type 1 - Plate or float, flat, quality q3.
Class 1 - Transparent.
Class 2 - Heat absorbing.
Style A - Higher light transmission.
Style B - Lower light transmission.

Class 3 - Light reducing, tinted.
Class 4 - Translucent

Type II - Sheet, flat, quality q6.
Class 1 - Transparent.
Class 2 - Heat absorbing.
Style A - Higher light transmission.
Style B - Lower light transmission.
Class 3 - Light reducing, tinted.
Class 4 - Translucent.

Type III - Rolled, flat.
Class 1 - Translucent.
Class 2 - Heat absorbing.
Style A - Higher light transmission.
Style B - Lower light transmission.
Class 3 - Light reducing, tinted.

Form 3 - Figured.
Quality q10 - Decorative.
Quality q11 - Glazing.

Finish f1 - Figured one side.
Finish f2 - Figured both sides.

Style 1 - Uncoated surfaces
Style 11 - One surface ceramic coated.
Style III - Coated surface, transparent coating.

Type - 1 - Primary glass (float or sheet).
Quality q3 - Glazing select.
Quality q5 - Glazing B.
Class 1 - Transparent.
Class 2 - Heat absorbing and light reducing.
Kind A - Higher light transmittance.

Kind B - Lower light transmittance.
Class 3 - Light reducing, tinted.
Class 4 - Translucent.
Type II - Rolled flat.
Quality q7 - Decorative.
Quality q8 - Glazing.
Class 1 - Translucent.

Class 2 - Heat absorbing.
Kind A - Higher light transmittance.
Kind B - Lower light transmittance.
Class 3 - Light reducing, tinted.
Form 3 - Figured
Finish 1 - Figured one side.
Finish 2 - Figured both sides.
Pattern p1 - Linear.
Pattern p2 - Geometric
Pattern p3 - Random.
Pattern p4 - Special.
Pattern P12 - Custom designs or textures.

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Commanding Officer

Form 4 - Rough surface finishes (rolled rough plate).
  Finish f3 - Rough one side, polished other.
  Finish f4 - Rough one side, smooth other.
  Finish f5 - Rough both sides.

Condition B, spandrel glass:

Type I - Plate or float, flat, quality q3.

Type II - Sheet, flat, quality q6.
  Form 3 - Figured.
    Quality q7 - Decorative.

Type III - Rolled, flat.
  Form 3 - Figured.
  Quality q10 - Decorative.
  Quality q11 - Glazing.
    Finish f6 - Figured one side, smooth one side, either side ceramic coated.
    Form 4 - Rough surface finishes (rolled rough plate).
    Finish f7 - Rough one side, polished other side, polished side ceramic coated.
    Finish f8 - Rough one side, smooth other side, smooth side ceramic coated.
    Finish F9 - Rough both sides, one side ceramic coated.

Patterns, type III, form 3:

Pattern p1 - Linear, uniform.
  p2 - Linear, random.
  p3 - Geometric, deep.
  p4 - Geometric, shallow.
  p5 - Random, deep.
  p6 - Random, shallow.
  p7 - Fluted, large.
  p8 - Fluted, small.
  p9 - Shallow, uniform, quality oil.
  p10 - Spaced ribs, quality q11.
  p11 - Special.

Style II, one surface ceramic coated:

Type I - Primary glass (float or sheet).
  Quality q3 - Glazing select.
  Quality q5 - Glazing B.

Type II - Rolled, flat.
  Quality q8 - Glazing.
  Finish f6 - Finish f1 with either side ceramic coated.

Type III - Style III glass.

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Pattern, types, I, II, or III:

Pattern p12 - Custom designs or textures.

Style III, coated surface, transparent coating.
Type 1 - Primary glass (float or sheet).
Quality q3 - Glazing select.
Class 1 - Transparent.
Class 2 - Heat absorbing and light reducing.
Style A - Higher light transmission.
Style B - Lower light transmission.
Class 3 - Light reducing, tinted.

MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review Activities

Air Force - 84

User Activities

Army - CE
Navy - MC

PREPARING ACTIVITY:

Navy - YD

Project No. 5620-0074

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4.6.6 Chemical penetration. Following the general procedures in accordance